

1) Shunt Harmonic Filter Calculations

A shunt harmonic filter (series RLC) is shown in Fig. 1(a). It is designed for a three-phase 13.8 kV, 60 Hz system to shunt the 7th harmonic current. The circuit parameters are:

- $r_L = 1 \Omega$
- $L = 19 \text{ mH}$
- $C = 8.2 \mu\text{F}$

Step 1: Write the KVL equation:

$$V_{dc} - r_L i_L - L \frac{di_L}{dt} - v_C = 0$$

Or equivalently:

$$L \frac{di_L}{dt} + r_L i_L + v_C = V_{dc}$$

With $i_L = C \frac{dv_C}{dt}$ and $v_C = \frac{1}{C} \int i_L dt$.

Step 2: Standard second-order ODE:

$$\frac{d^2 i_L}{dt^2} + \frac{r_L}{L} \frac{di_L}{dt} + \frac{1}{LC} i_L = 0$$

Step 3: Characteristic equation:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

Step 4: Calculate parameters:

$$\omega_n = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{19 \times 10^{-3} \times 8.2 \times 10^{-6}}} = 2533.5 \text{ rad/s}$$

$$\zeta = \frac{r_L}{2\sqrt{\frac{L}{C}}} = \frac{1}{2\sqrt{\frac{19 \times 10^{-3}}{8.2 \times 10^{-6}}}} = 0.0104$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} = 2533.3 \text{ rad/s}$$

Step 5: Quality factor:

$$Q_{\text{factor}} = \frac{\omega_n L}{r_L} = \frac{2533.5 \times 0.019}{1} = 48.136$$

Step 6: Eigenvalues:

$$\lambda = -\zeta\omega_n \pm j\omega_d = -26.316 \pm 2533.3j$$

Step 7: Reactive power:

$$Q_{\text{kVAr}} = -602.04 \text{ kVAr}$$

All calculations confirm the system is very underdamped ($\zeta = 0.0104$).

Summary of Key Results:

- Natural frequency: $\omega_n = 2533.5 \text{ rad/s}$
- Damping ratio: $\zeta = 0.0104$
- Damped frequency: $\omega_d = 2533.3 \text{ rad/s}$
- Quality factor: $Q = 48.136$
- Eigenvalues: $-26.316 \pm 2533.3j$
- Reactive power: -602.04 kVAr

$$x(t) = u_0 \left[1 - \frac{1}{\sqrt{1 - \zeta^2}} e^{-\zeta \omega_n t} \sin(\omega_d t + \phi) \right]$$

$$x(t) = 11.3 \left[1 - 1.00005 e^{-26.34 t} \sin(2533.3 t + 1.5604) \right]$$